Python Pandas

Pandas is a powerful Python library used for working with data sets. It provides functions for:

1. **Analyzing**
2. **Cleaning**
3. **Exploring**
4. **Manipulating**

The name "Pandas" references both "Panel Data" and "Python Data Analysis" and was created by Wes McKinney in 2008.

Pandas allows us to analyze big data and make conclusions based on statistical theories. It can clean messy data sets, making them readable and relevant. Relevant data is crucial in data science.
Usually, we work with two types of datasets:

- **Series**
- **DataFrame**

1.Data Reading

2.Data Filtering

- Data Frame Filtering
- Conditional Filtering
- Single Value Filtering

13.Data Exploration

.head()
.tail()
.info()
.describe()
.value_counts()
.unique()
.isnull().sum()
.notnull().sum()

3.Handling Missing Values

.fillna
.dropna

4.Adding New Columns

5.Replacing/Updating Values

- replace
- update

6.Dropping Duplicates

7.Drop Rows/Columns

8.Changing Column Names

10.Working with Index

11.Sorting

9. Iteration Over Data Frame

- Iterrows()
- Itertuples()
- Items()

10. Use of

- Apply
- Map
- applymap

12.Joining Data frames

- Concatenation
- Joining
- Merging

14.Pivot Function

15.Group By

16.String Handling